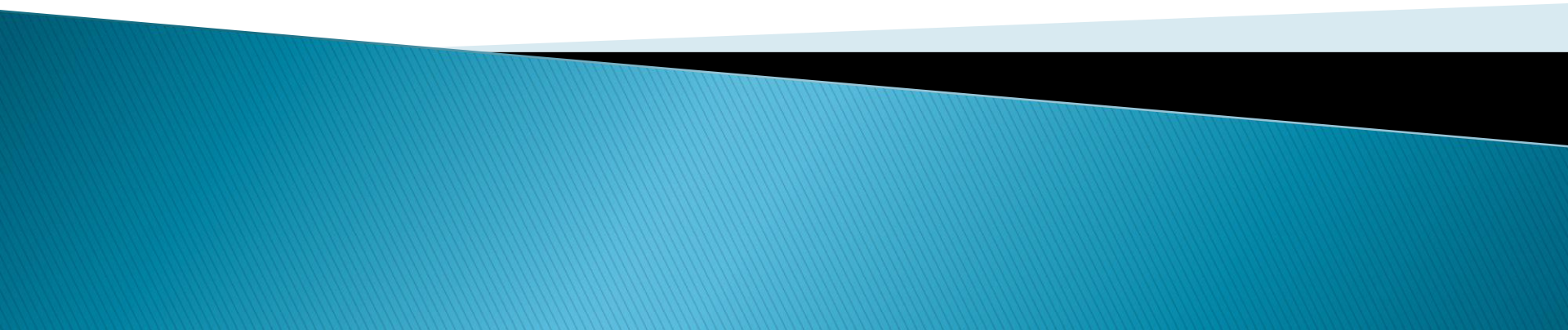


EC5001

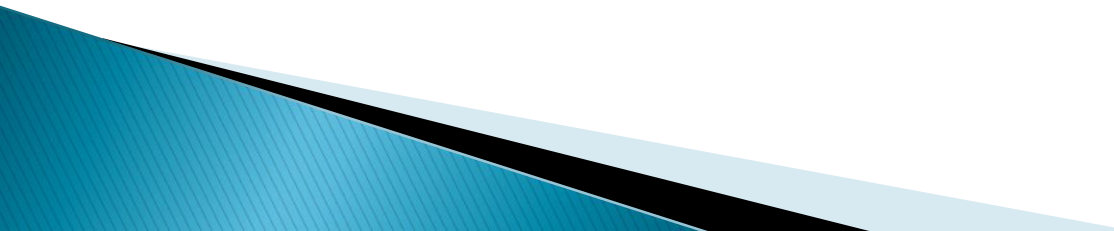
Introduction to EC

Lecture 5

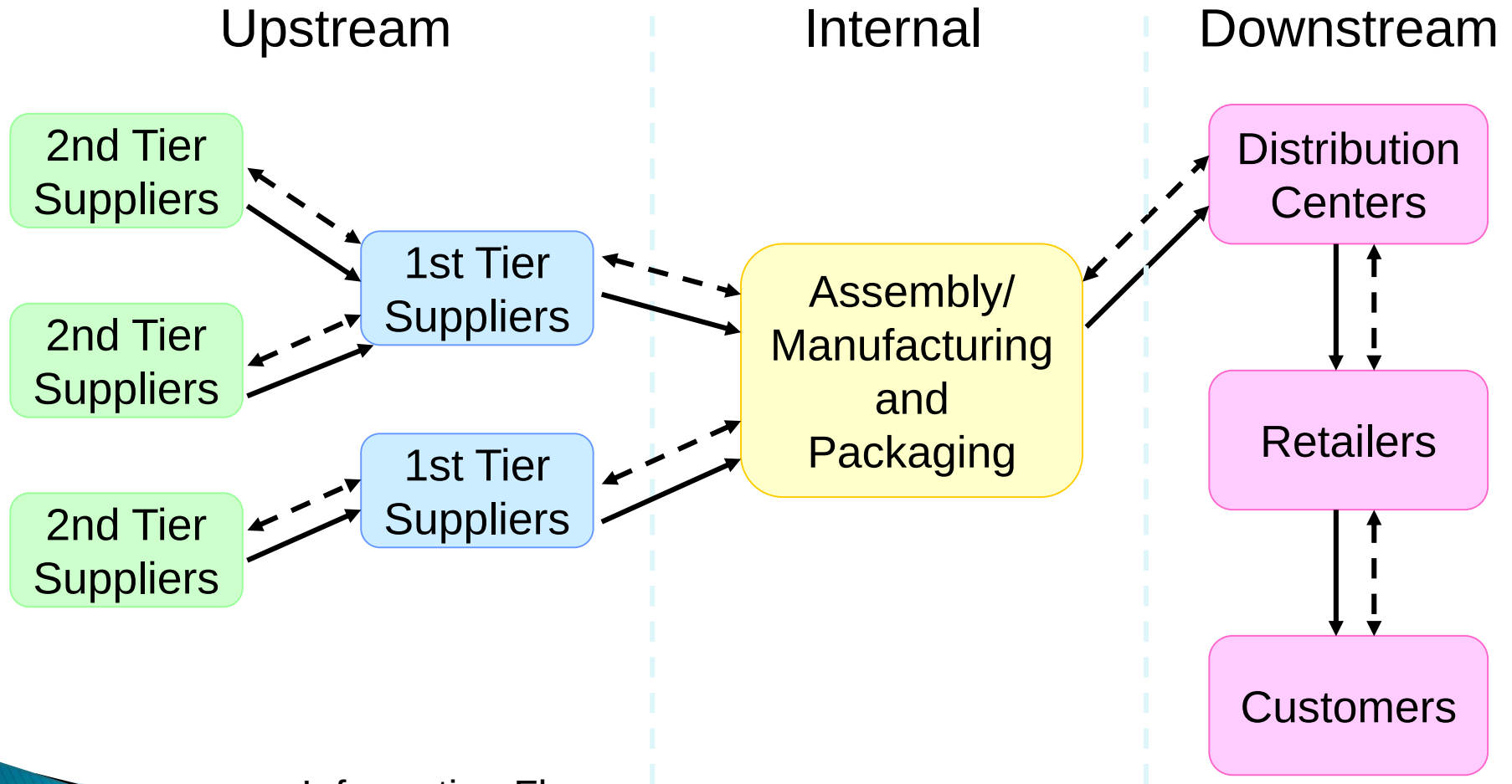
M-Commerce & E-business plan



Recap – What is a supply chain?

- ▶ A supply chain is a collection of companies and processes moving a product:
 - From suppliers of raw materials to
 - suppliers of intermediate components, to
 - final production,
 - to the customer
 - ▶ **Upstream**—flow from sources of raw materials and components
 - ▶ **Downstream**—flow to customers
 - ▶ Suppliers have their own supply chain
 - A better name: supply network
- 

Recap – Supply Chain: Generic Model



Key: Information Flow
 Physical/ Material Flow

Recap – Business-to-Business E-commerce

- ▶ B2B involves exchanges between two or more businesses
 - B2B does not include end customers (consumers)
 - Transactions in supply networks
- **Business-to-Business e-commerce (B2B EC)**

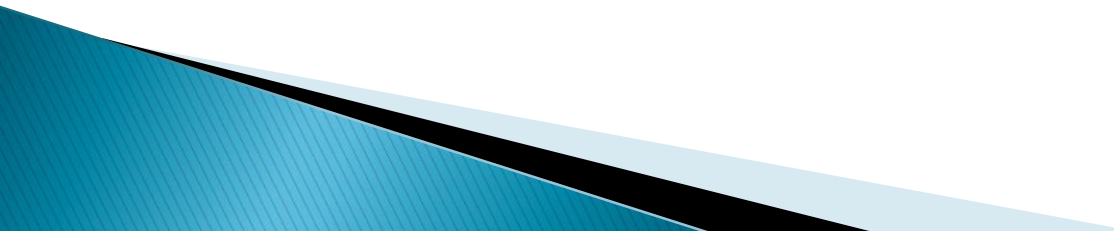
Transactions between businesses conducted electronically over the Internet, extranets, intranets, or private networks; also known as eB2B (electronic B2B) or just B2B

Recap – Concepts & characteristics of B2B EC

▶ B2B EC Characteristics

- Parties to the transaction
 - Buyers
 - Sellers
 - Online intermediaries

▶ Types of B2B transactions

- Spot buying
 - Strategic sourcing
- 

Recap – Concepts & characteristics of B2B EC

- **Vertical marketplaces**

Markets that deal with one industry or industry segment (e.g., steel, chemicals)

- **Horizontal marketplaces**

Markets that concentrate on a service, material, or a product that is used in all types of industries (e.g., office supplies, PCs)

Recap – B2B Portals

- **Corporate portal**

A major gateway through which employees, business partners, and the public can enter a corporate website

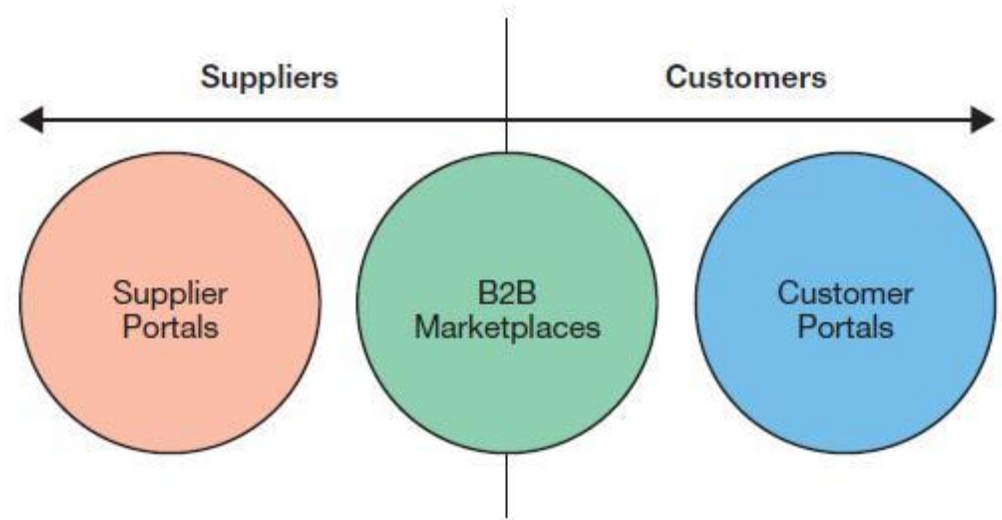
- **Examples**

- **Types of Corporate Portals**

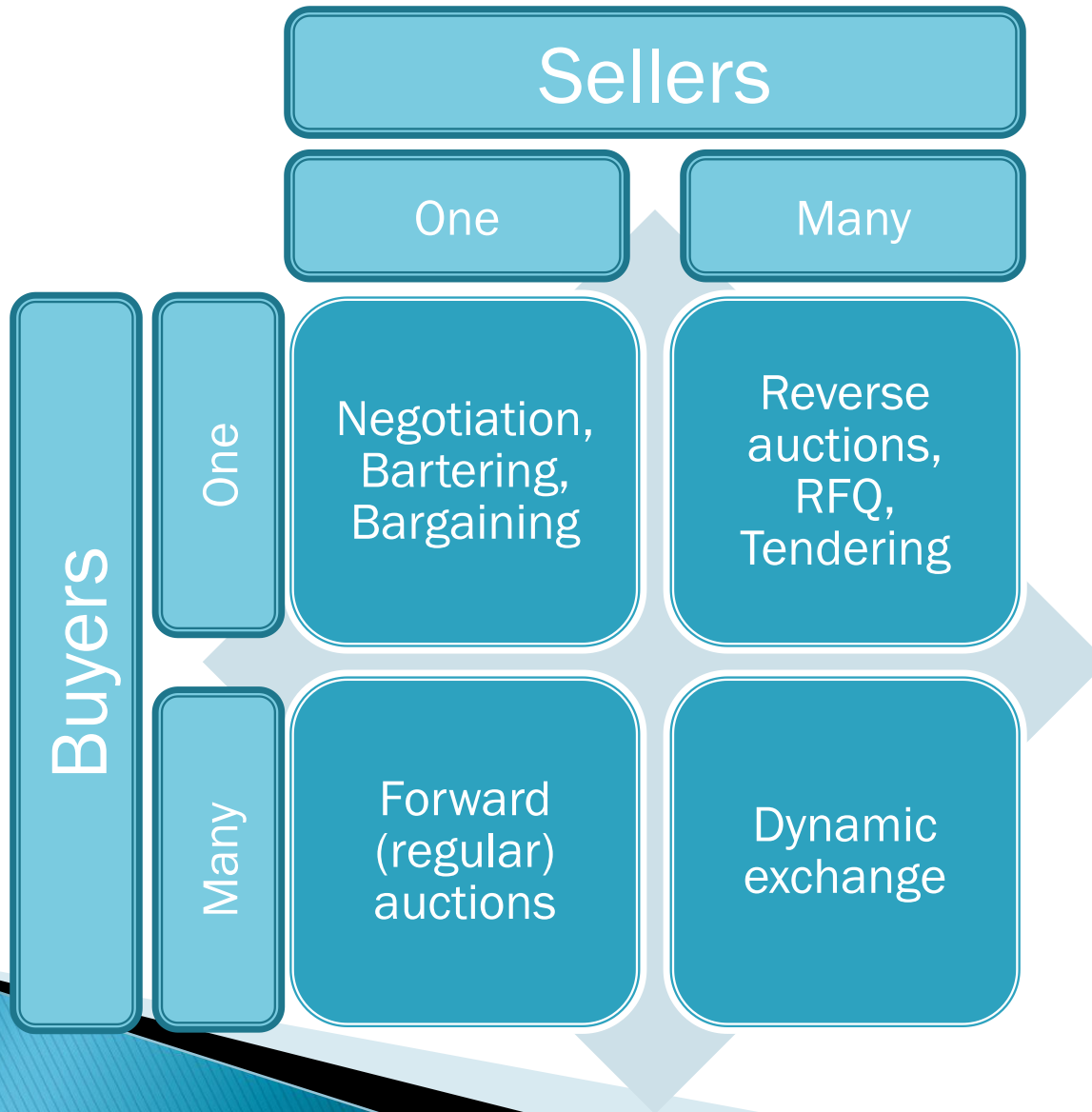
- Portals for Suppliers and Other Partners
 - Customer Portals
 - Employee Portals
 - Executive and Supervisor Portals
 - **mobile portals**

Recap – B2B Marketplaces

- ▶ B2B Marketplaces / B2B exchanges
 - Operated by third-party vendors
 - Allow many buyers and many suppliers to interact
 - Some operate within a vertical market (industry-specific); for example:
 - steellink.com
 - paperindex.com
 - fibre2fashion.com
 - Some are not industry-specific
 - Alibaba.com



Recap – B2B – Electronic auctions



Recap – Primary Threats to Security

- ▶ Intentional or unintentional
- ▶ Natural disasters
 - Power outages, hurricanes, floods, and so on
- ▶ Accidents
- ▶ Employees and consultants
- ▶ Links to outside business contacts
 - Travel between business affiliates
- ▶ Outsiders
 - Viruses
- ▶ Company policies



Recap – Managing IS Risk

- ▶ Any information system can be compromised
- ▶ Organizations need to manage risks
- ▶ Three components:
 - **Threats:** undesirable events such as fire, software error, hacking, computer viruses
 - **Vulnerabilities:** weaknesses that can be exploited by threats to cause damage
 - **Impacts:** severity of the consequence(s) of a threat exploiting a vulnerability and resulting in damage

Recap – Computer Crime

- ▶ Computer crime—The act of using a computer to commit an illegal act
 - Targeting a computer while committing an offense
 - Using a computer to commit an offense
 - Using computers to support a criminal activity
- ▶ Overall trend for computer crime has been declining over the past several years
 - However: many incidents are never reported
 - Reputation, etc.

Recap – Computer Viruses and Other Destructive Code

▶ Malware

- short for “malicious software” such as viruses, worms, and Trojan horses

▶ Virus

- Destructive program that disrupts the normal functioning of computer software

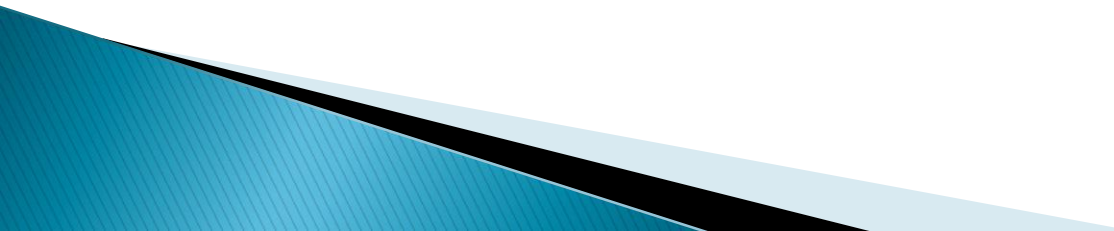
▶ Worm

- Variation of a virus that is targeted at networks
- Designed to spread by itself

▶ Trojan Horse

- Malicious program disguised as a desirable program
- Does not replicate
- E.g., keyloggers disguised as games

Recap – Other Threats

- ▶ Unauthorized Access
 - ▶ Information Modification
 - ▶ Industrial espionage
 - ▶ Spyware
 - ▶ Phishing
 - ▶ Internet Hoaxes
 - ▶ Identity Theft
- 

Recap – Software Piracy

- ▶ Legal activities
 - Making one backup copy for personal use
 - Sharing free software (shareware or public domain software)
- ▶ Illegal activities
 - Making copies of purchased software for others
 - Offering stolen proprietary software (warez peddling)
- ▶ Intellectual property
 - Patents: process or machine inventions
 - Copyrights: creations of the mind
 - Various copyright laws applicable to software

Recap – Technological Safeguards

- ▶ Physical access restrictions
- ▶ Firewalls
- ▶ Encryption
- ▶ Virus Monitoring and prevention
- ▶ Audit–control software
- ▶ Dedicated facilities

Mobile Commerce

Drivers of M-Commerce :

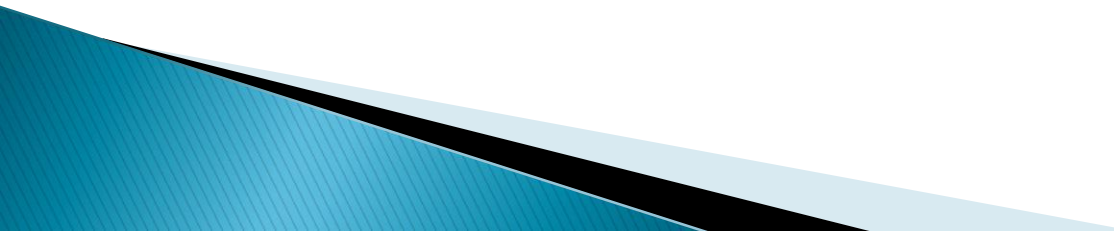
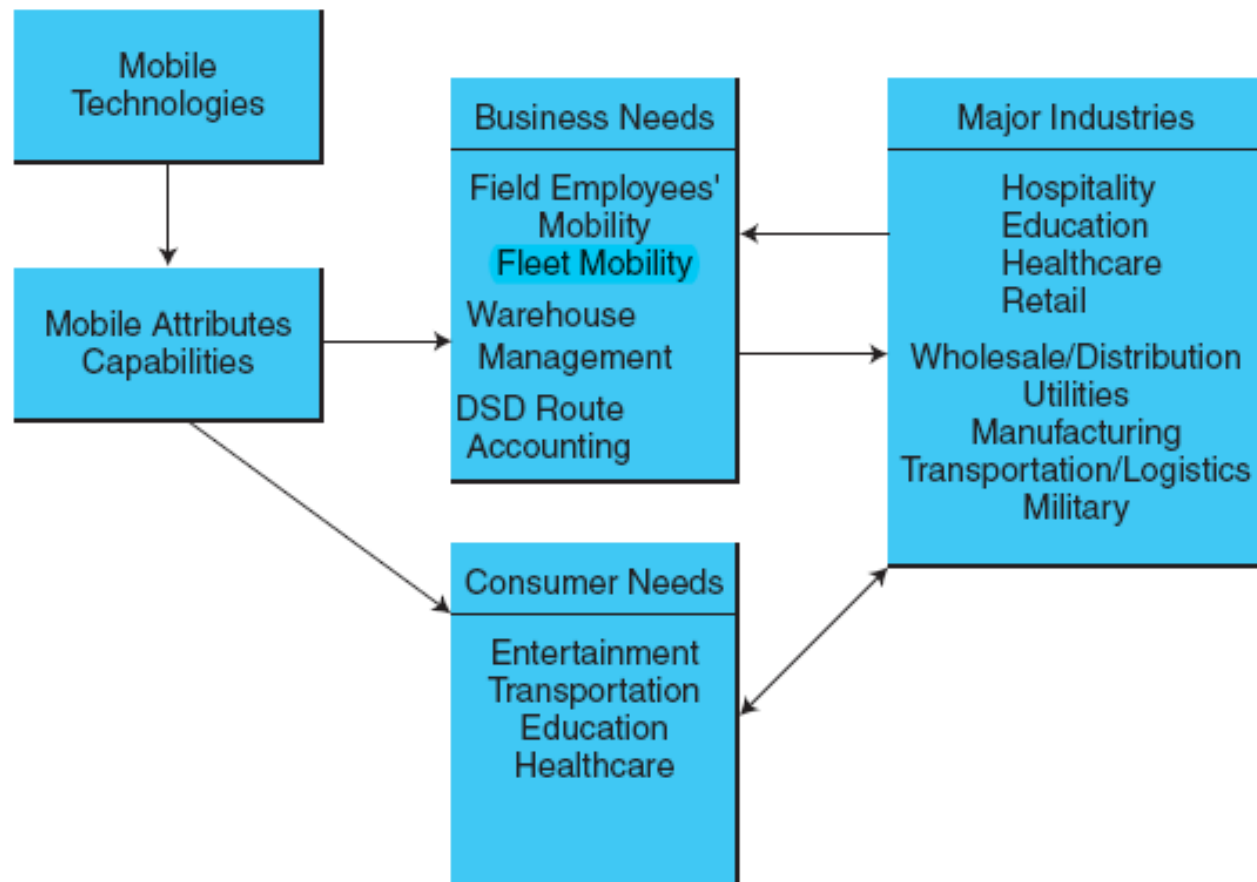
- Widespread availability of more powerful mobile devices
 - The handset culture
 - The service economy
 - Vendor's push
 - The mobile workforce and mobile enterprise
 - Improved price/performance
 - Improving bandwidth
- 

EXHIBIT 6.2

M-Commerce Applications and Their Classifications



Technical Infrastructure: Mobile Computing

Overview of Mobile Computing :

- **Wireless mobile computing (mobile computing)**
Computing that connects a mobile device to a network or another computing device, anytime, anywhere

EXHIBIT 6.3

Mobile Computing Basic Terminology

Bluetooth. A chip technology wireless standard designed for temporary, short-range connection (data and voice) among mobile devices and/or other devices (see bluetooth.org).

Global Positioning System (GPS). A satellite-based tracking system that enables the determination of a GPS device's location. (See Section 6.6 for more on GPS.)

Personal Digital Assistant (PDA). A small portable computer, such as BlackBerry handhelds and the pocket PC devices from companies like Research In Motion or Palm.

Short Messaging Service (SMS). A technology for sending short text messages (up to 160 characters) on cell phones. SMS messages can be sent or received concurrently, even during a voice or data call. Used by hundreds of millions of users, SMS is known as "the e-mail of m-commerce." Some companies offer multilanguage text creation.

Smartphones. Internet-enabled cell phones that can support mobile applications. These "phones with a brain" are becoming standard devices. They include WAP microprocessors for Internet access and the capabilities of PDAs as well. The iPhone is the most popular example of a smartphone.

WiMAX. A wireless technology based on the IEEE 802.16-2004 standard, designed to provide Internet access across metro areas to fixed (not moving) users. It is considered wireless broadband technology.

Wireless Application Protocol (WAP). A technology that offers Internet browsing from wireless devices.

Wireless Local Area Network (WLAN). A broad term for all 802.11 standards. Basically, it is a wireless version of the Ethernet networking standard.

For an extensive list of other terms, see harvest.cals.ncsu.edu/index.cfm?showpage=291 and webopedia.com/Mobile_Computing.

Technical Infrastructure: Mobile Computing

Mobile Devices:

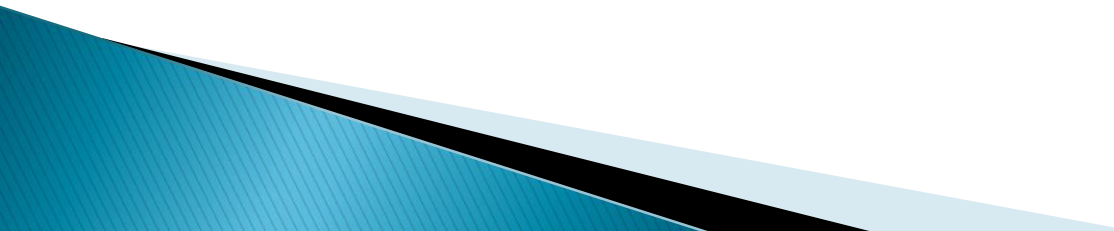
- **Personal digital assistant (PDA)**

A stand-alone handheld computer principally used for personal information management

Smartphone

A mobile phone with PC-like capabilities

Tablets



Technical Infrastructure: Mobile Computing

Other Mobile Devices

- ▶ Smartbooks
- ▶ Wearable devices
- ▶ Camera
- ▶ Touch-panel display
- ▶ Keyboard
- ▶ Speech translator
- ▶ Watch-like device
- ▶ RFID (radio frequency identification)
- ▶ Scanners

Technical Infrastructure: Mobile Computing

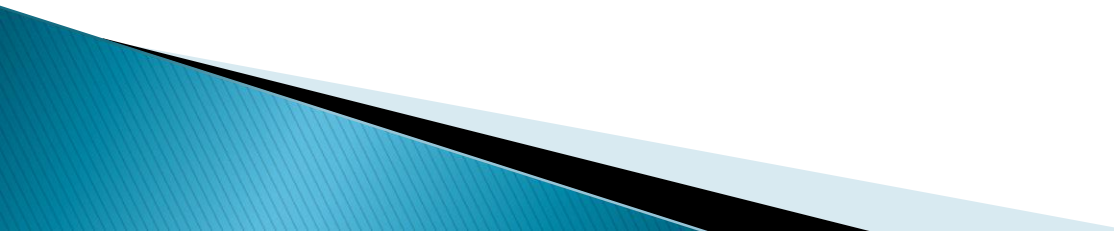
Mobile portal

A gateway to the Internet optimized for mobility that aggregates and provides content and services for mobile users

Content providers

Short message service (SMS)

A service that supports the sending and receiving of short text messages on mobile phones



Technical Infrastructure: Mobile Computing

Multimedia messaging service (MMS)

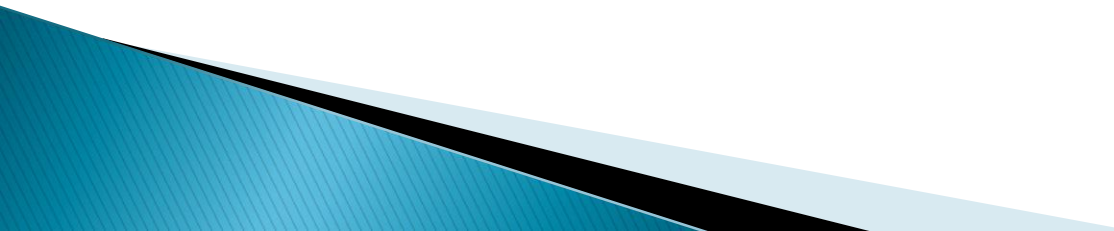
The emerging generation of wireless messaging; MMS is able to deliver rich media

Location-Based Services

Voice-Support Services

- **interactive voice response (IVR)**

A voice system that enables users to request and receive information and to enter and change data through a telephone to a computerized system



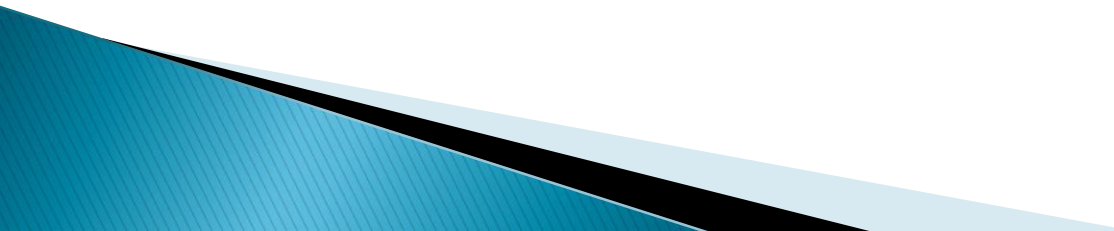
Technical Infrastructure: Mobile Computing

Personal area network (PAN)

A wireless telecommunications network for device-to-device connections within a very short range

Bluetooth

A set of telecommunications standards that enables wireless devices to communicate with each other over short distances



Technical Infrastructure: Mobile Computing

Wireless local area network (WLAN)

A telecommunications network that enables users to make short-range wireless connections to the Internet or another network

Wi-Fi (wireless fidelity)

The common name used to describe the IEEE 802.11 standard used on most WLANs

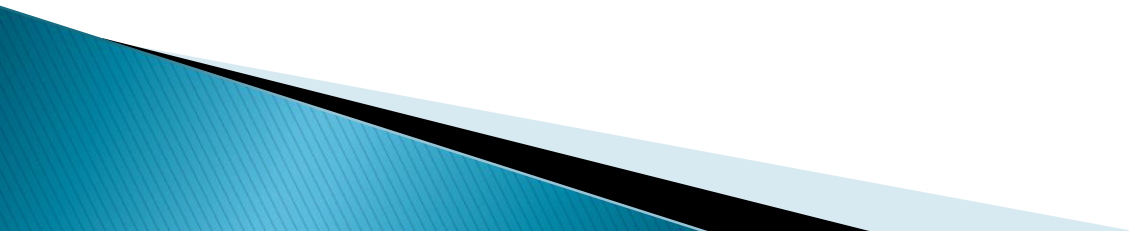
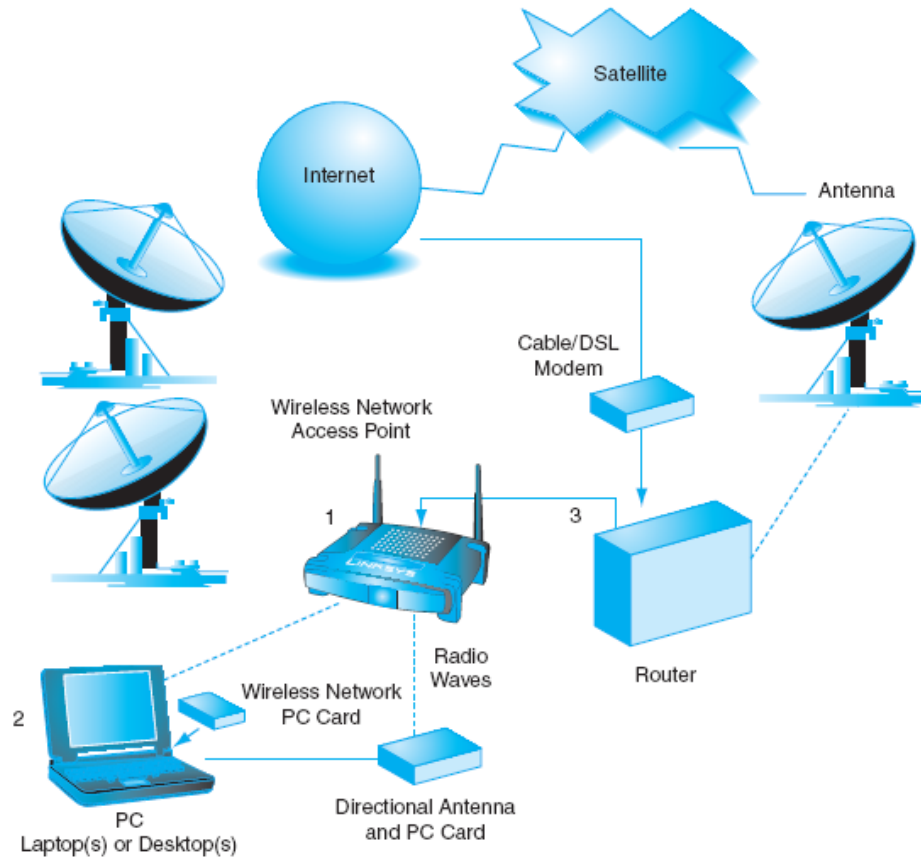


EXHIBIT 6.4 How Wi-Fi Works



- 1 Radio-equipped access point connected to the Internet (or via a router). It generates and receives radio waves (up to 400 feet).
- 2 Several client devices, equipped with PC cards, generate and receive radio waves.
- 3 Router is connected to the Internet via a cable or DSL modem or connected via a satellite.

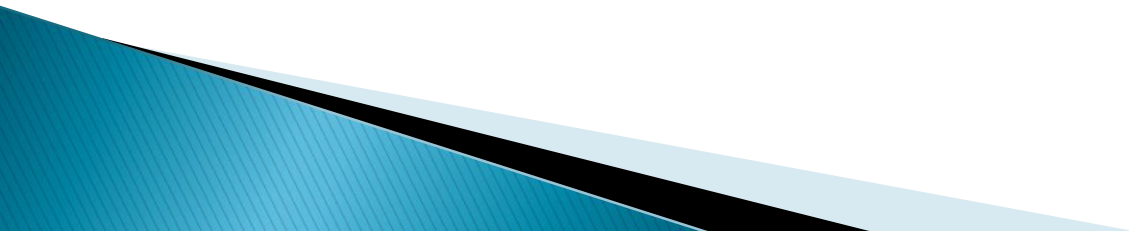
M- Commerce Applications

- ▶ **Mobile banking**

Performing banking activities such as balance checks, account transactions, payments, credit applications, etc., via a mobile device

- ▶ **Mobile Stock Trading**

- ▶ **Real Estate**



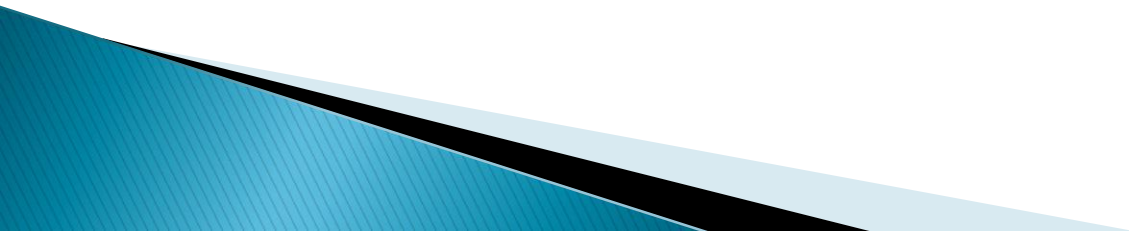
M- Commerce Applications

- ▶ **Mobile entertainment**

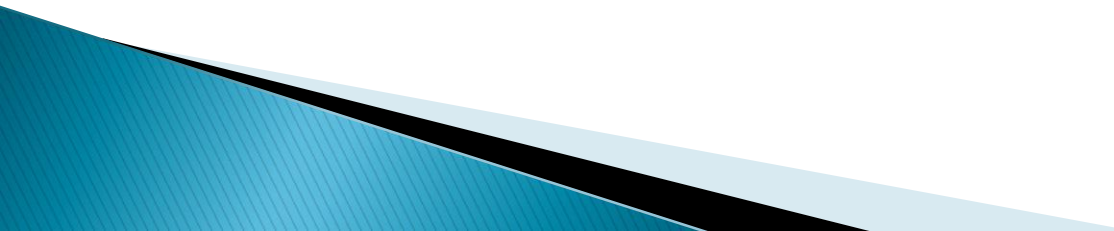
Any type of leisure activity that utilizes wireless telecommunication networks, interacts with service providers, and incurs a cost upon usage

- ▶ **Mobile Games**

- ▶ **Mobile Gambling**



Location-Based M-Commerce

- ▶ **Location-based m-commerce (l-commerce)**
Delivery of m-commerce transactions to individuals in a specific location, at a specific time
 - ▶ **Real-time location system (RTLS)**
Systems used to track and identify the location of objects in real time
- 

Location-Based M-Commerce

- ▶ **Geolocation**

The process of automatically identifying a Web user's physical location without that user having to provide any information

- ▶ **Network-based positioning**

Relies on base stations to find the location of a mobile device sending a signal or sensed by the network

Location-Based M-Commerce

- ▶ **Terminal-based positioning**

Calculating the location of a mobile device from signals sent by the device to base stations

- ▶ **Global positioning system (GPS)**

A worldwide satellite-based tracking system that enables users to determine their position anywhere on the earth

Location-Based M-Commerce

- **Location-Based Data**

- Locating
- Navigating
- Searching
- Identifying
- Event checking

- **Geographical information system (GIS)**

A computer system capable of integrating, storing, editing, analyzing, sharing, and displaying geographically referenced (spatial) information



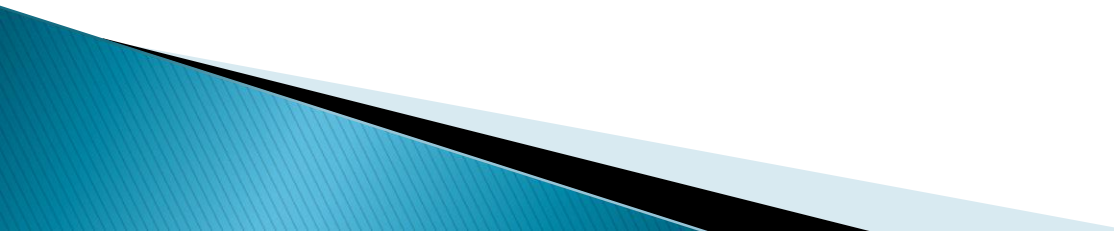
Location-Based M-Commerce

▶ Location-based service (LBS)

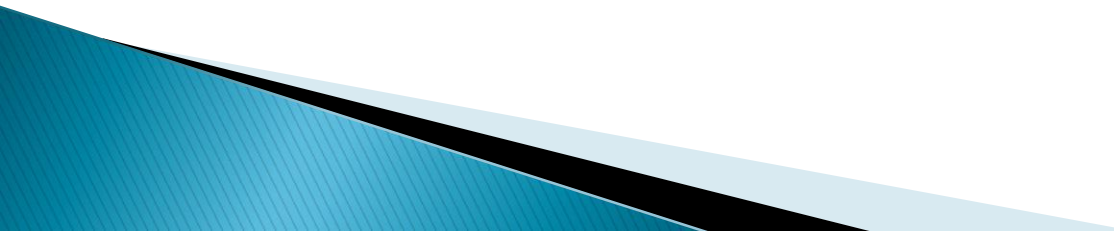
An information service accessible from and to mobile devices through a mobile network utilizing the ability to make use of the geographical position of the mobile device to deliver a service to the user

- Automatic vehicle location (AVL)

Social location-based marketing

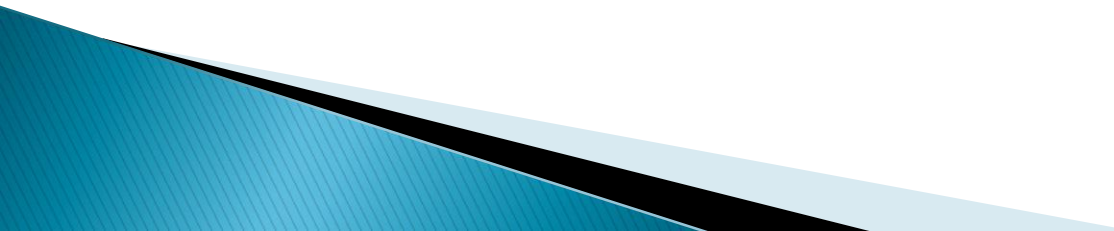


Business Plan

- ▶ What is a business plan?
 - ▶ A written document that identifies a company's goals and outlines how the company intends to achieve those goals
 - ▶ Important for starting a new business or applying for business loans
 - ▶ Also vital for running a business
 - new loans or new investments
 - plans to optimize growth and development according to priorities
- 

Business Plan

- ▶ Why write a business plan?

- (1) To acquire funding;
 - (2) To acquire other resources;
 - (3) To recruit senior management;
 - (4) To make you a better business owner;
 - (5) To make a realistic approach to your business;
 - (6) To decide not to develop the business;
 - (7) To keep you on track.
- 

Business Plan

- ▶ Elements of a business plan
 - Executive summary
 - Company overview & mission statement
 - Description of product/service
 - Market analysis
 - Marketing plan
 - Implementation strategy/action plan
 - Financial review
 - Risks and challenges

Business Plan

- ▶ Presenting an e-business plan
- ▶ Format depends on presentation context
- ▶ Have more than one format for the same business plan:
 - An “elevator pitch”
 - An oral presentation
 - A written presentation for external stakeholders

Executive summary

- ▶ May be the most important part of business plan
 - The first part everyone will read
 - May be the *only* part that some individuals will read
- ▶ Synopsis of the key points of the entire e-business plan (1–2 pages)
 - To explain the fundamentals of the business in a way that both informs and excites the reader
- ▶ An outline of what is contained in the full plan document
 - Providing enough information for readers to understand the key points, what is unique about the business and why the business will be successful
- ▶ When to write?
 - Start in the middle of the plan-writing process and finish it last

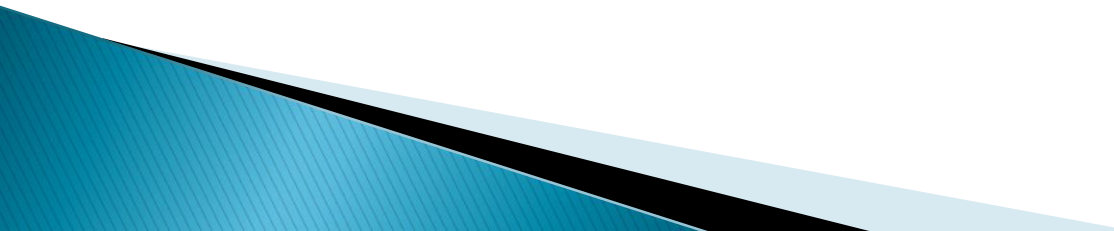
Company overview

- ▶ What is your company all about?
 - ▶ Mission statement
 - An opportunity to define your business at the most basic level
 - Should tell your company story and ideals in less than 30 seconds
 - who your company is, what you do, what you stand for, and why you do it
 - Fundamental, unique purpose that sets a business apart from other firms of its type
 - Identifies the scope of the business' operations in product and market terms
- 

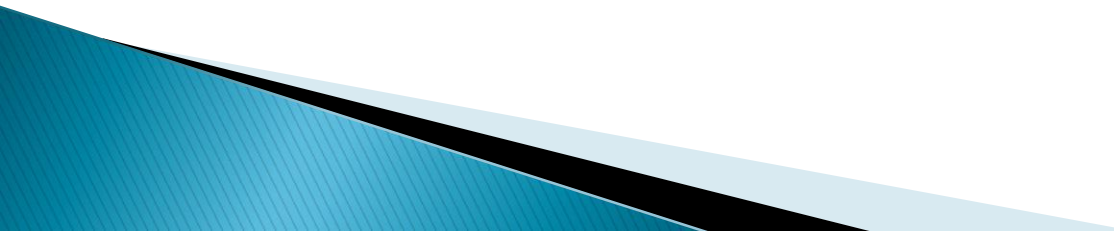
Product and services

- ▶ Description of the products or services the business will provide to customers
- ▶ Should be complete enough to give the reader a clear idea of your products/services
 - Not overly detailed or technical so the reader is confused or loses interest and attention

Product and service

- ▶ Tell the reader what makes your company's product or service different or unique in the marketplace!
 - ▶ Briefly highlight product distinctions and key selling attributes
 - ▶ What benefits will the customer receive from purchasing the product or service?
- 

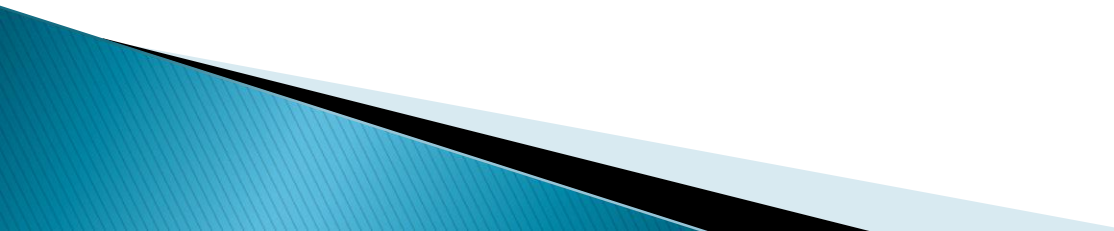
Market analysis

- ▶ Demonstrates that you know your market
 - Environmental analysis
 - ▶ Demonstrates that you know your customers
 - Who they are, their characteristics, and why they are likely to buy from your business
 - ▶ Effective marketing begins with careful, systematic research
 - Don't assume that you already know about your intended market
- 

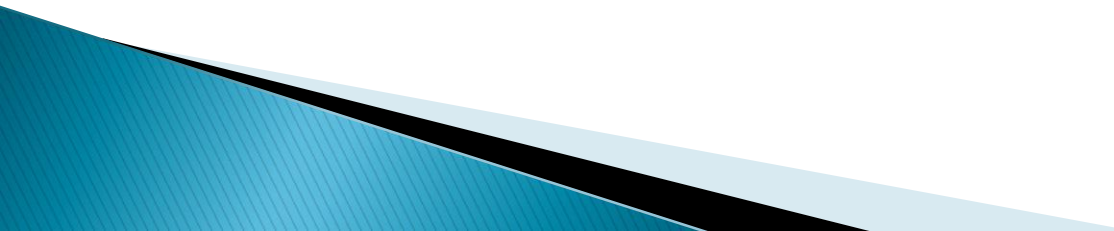
Market analysis

- ▶ Potential market
 - total number of people who could buy from your business
- ▶ Addressable market
 - the group of individuals in the potential market who are likely to have an interest in what your business has to offer
- ▶ Target market
 - the group(s) of individuals in the addressable market that are likely to buy from your website

Market analysis

- ▶ Define your primary, secondary, and, perhaps, tertiary target markets
 - ▶ Target markets are based on segmentation characteristics within the total addressable market
 - Demographic characteristics (age, gender, income)
 - Geographic characteristics (state, nation, country, city, suburb)
 - Psychographic characteristics (lifestyle, personality, values)
 - Consumer characteristics (loyalty, shopping frequency)
- 

Market analysis

- ▶ Market research
 - ▶ Gather concrete data about target markets
 - What are the specific demographics (age, education, income,) of these markets?
 - How many individuals are in the target market?
 - Do they use the Internet?
 - Do they buy online?
 - Are these markets growing? Stable? Declining?
- 

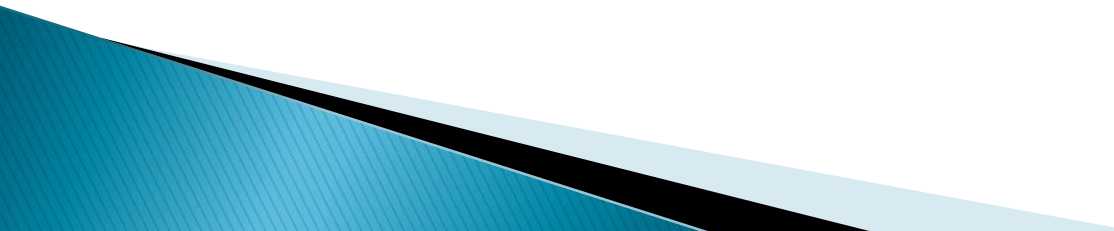
Competitor analysis

- ▶ Important requirement in any business plan because
 - It reveals the firm's competitive position in the market
 - It helps you to develop strategies to be competitive
 - Investors and other readers of the business plan will expect it

Competitor analysis

- ▶ Direct competitors
 - ▶ Indirect competitors
 - ▶ Future competitors

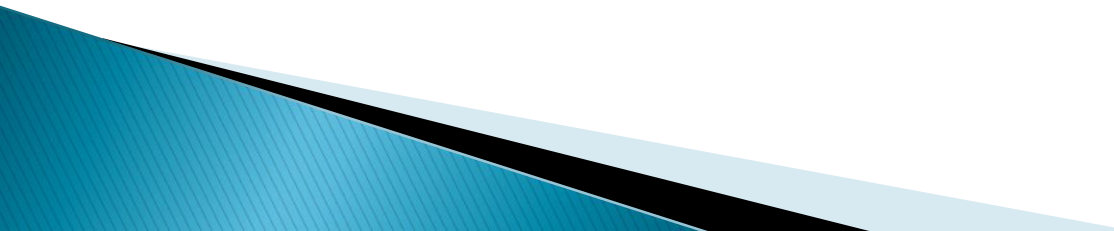
 - ▶ What are their strengths/weaknesses?
 - ▶ What are their sources of competitive advantage?

 - ▶ What is your competitive strategy?
- 

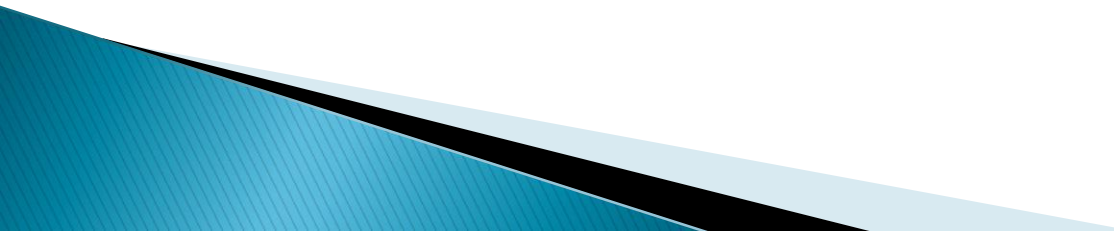
Sales and promotion

- ▶ How will you deliver your selling proposition to your prospective customers?
 - Communication strategy
 - Distribution strategy

Implementation strategy

- ▶ How does this web site (or app) achieve your business plan's objectives?
 - ▶ What will you include?
 - ▶ What is your development plan?
- 

Financial statements

- ▶ Projected results and future expectations
Revenue model
 - ▶ Start-up budget: details expected investment before the business starts to operate
 - ▶ Present breakeven analysis, payback period, net present value analysis, cash flow statement etc.
 - project the financial future of the company for the first 3 to 5 years
- 

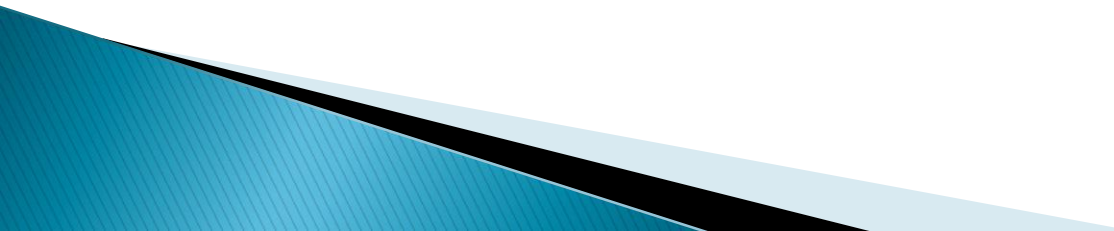
Risks and challenges

- ▶ Be open about any challenges you may face!

Finish your plan

- ▶ At the conclusion of this course you have all the major pieces you need to finish your e-business plan
 - ▶ Final steps:
 - Finish the executive summary
 - Write a conclusion
 - Put all the parts together in a single document and check for inconsistencies and duplication
 - Do all the formatting, spell-checking
 - Give the plan to a colleague to read and suggest changes
- 

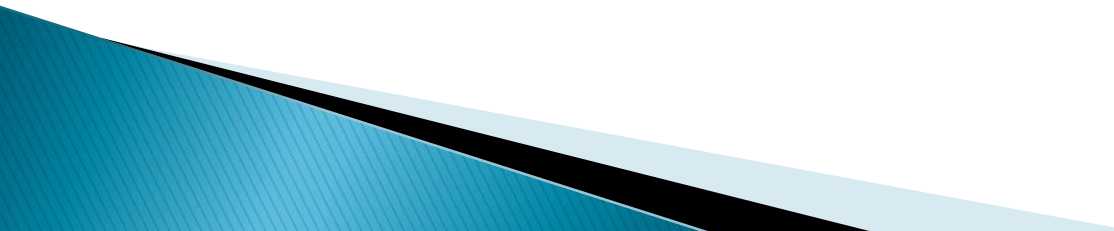
Finish your plan

- ▶ Make it attractive and easy to read
 - ▶ Do everything you can to make it easy for the reader!
 - ▶ Provide references for your claims
 - ▶ Use charts and graphs to make information easier to understand
- 

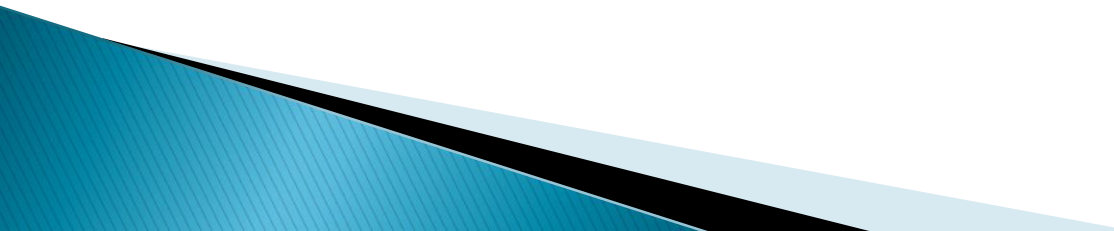


Business model canvas

Business model canvas

- ▶ Describing, analyzing, and designing business models
 - ▶ Business model: how an organization creates, delivers, and captures value
 - What does a company do?
 - How does a company uniquely do it?
 - In what way (or ways) does the company get paid for doing it?
 - What are the key resources and activities needed?
 - What are the costs involved?
- 

BMC—Customer segments

- ▶ The customers targeted with the product/service offering
 - Who will be our target customers?
 - Who are the most important customers?
 - ▶ Separate segments?
 - Needs require and justify a distinct offer
 - Reached through different distribution channels
 - Require different types of relationships
 - Have substantially different profitabilities
 - Are willing to pay for different aspects of the offer
 - ▶ Do you focus on mass market, niche markets?
- 

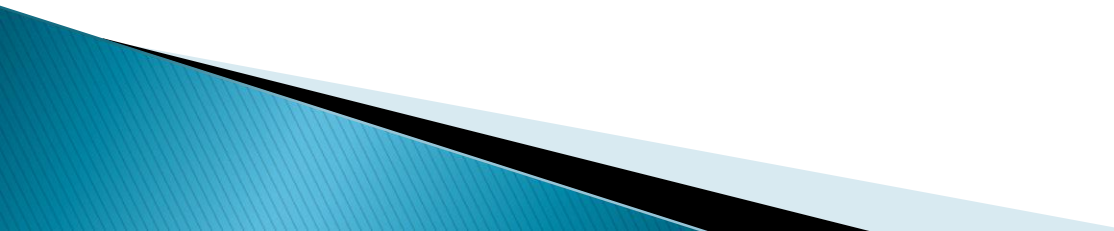
BMC—Value proposition

- ▶ The utility that the product/service has to offer to customers
 - Why do customers need our product/service?
 - What problems will our product/service solve?
 - Why would customers choose our product/service over our competitor's product/service?
- ▶ Newness, performance, design, brand/status, price, cost reduction, risk reduction, convenience

BMC—Channels

- ▶ The ways in which the product/service offerings reach the target customers
 - How will our customers be reached?
 - Which channels are best in terms of cost and convenience for the customers?
 - ▶ Channels needed for
 - Awareness
 - Evaluation
 - Purchase
 - Delivery
 - After sales
- 

BMC—Customer relationships

- ▶ The relationships formed with the target customers
 - What types of relationships do we build with our customers?
 - How do we maintain these relationships?
 - ▶ Needs:
 - Customer acquisition
 - Customer retention
 - Boosting sales (upselling)
 - ▶ Personal assistance, self service, automated services, communities, ...
- 

BMC—Revenue streams

- ▶ The way a firm generates income
 - One-time transaction revenues
 - Recurring revenues
 - ▶ Types:
 - Sales
 - Usage fees
 - Subscription fees
 - Renting/leasing
 - Licensing
 - Brokerage
 - Advertising
- 

BMC—Key resources

- ▶ The most important assets needed to make the business model work
 - What key resources are needed to enable our value proposition, channels, customer relationships, and revenue streams?
- ▶ Types of resources
 - Physical
 - Intellectual
 - Human
 - Financial

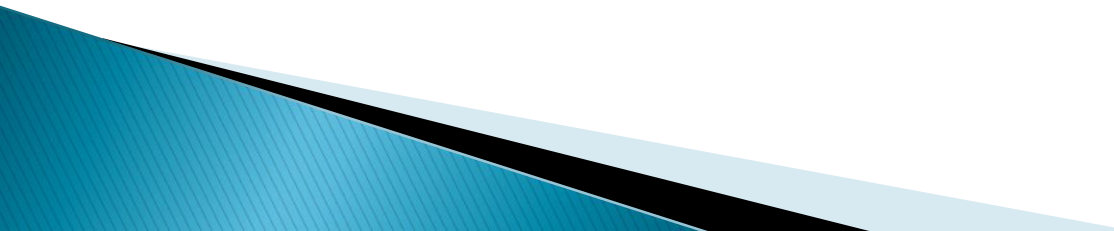
BMC—Key activities

- ▶ The most important activities needed to make the business model work
 - What key activities are needed to enable our value proposition, channels, customer relationships, and revenue streams?
- ▶ Types of activities
 - Production
 - Problem solving
 - Platform

BMC—Key partners

- ▶ The network of partners and suppliers needed to make the business model work
 - Who are our key partners and suppliers?
 - What resources do they offer, and what activities do they perform?
- ▶ Types of partners
 - Strategic alliances
 - Co-opetition
 - Joint ventures
 - Buyer-supplier relationships

BMC—Cost structure

- ▶ The costs incurred when operating the business model
 - What are the costs incurred when operating the business model?
 - Which resources and activities are most expensive?
 - ▶ Cost-driven vs. value-driven
 - ▶ Fixed costs vs. variable costs
 - ▶ Economies of scale vs. economies of scope
- 

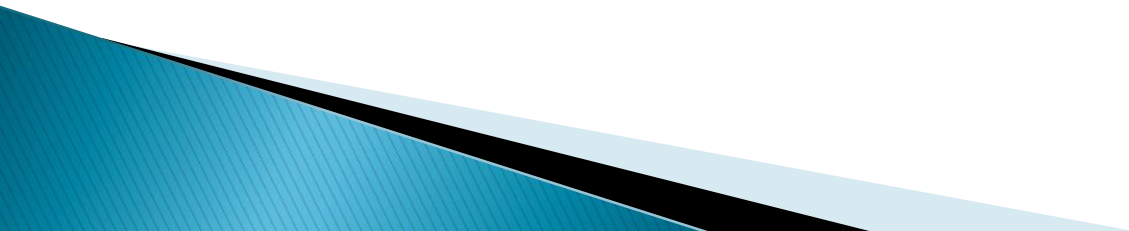
In Class Discussion

Business Model Canvas (BMC)

In Class Discussion

- ▶ Business Model Canvas (BMC) for the following companies
- ▶ Amazon
- ▶ Apple

Use the BMC template for the exercise.



Business Model Canvas – Amazon

 Key Partners	 Key Activities	 Value Propositions Write directly to the canvas... Or use the post-it™ note Double click on the post-it™ to edit. Recolour it using the picture format tools.	 Customer Relationships	 Customer Segments
	 Key Resources		 Channels	
 Cost Structure	 Revenue Streams			

Business Model Canvas – Apple

 Key Partners	 Key Activities	 Value Propositions Write directly to the canvas... Or use the post-it™ note Double click on the post-it™ to edit. Recolour it using the picture format tools.	 Customer Relationships	 Customer Segments
	 Key Resources		 Channels	
 Cost Structure	 Revenue Streams			

Next week – Interim Presentation

- ▶ Each team should submit a powerpoint presentation file on Canvas on Week 6.
- ▶ **Company background** (History—if any, mission and future strategy)
- ▶ **Market** (competitive analysis – Porter's Five Forces/SWOT/PEST, market segmentation, market demands, market growth)
- ▶ **Products/Services** (Your product or service and what makes it special and attractive; what is your e-business concept)